

Adaptive Spatial Filtering and Linear Discriminant Learning for Robust Motor Imagery EEG Classification

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Abstract—This study explores the use of electroencephalography (EEG) signals for classifying motor imagery (MI) tasks involving imagined left- and right-hand movements. To achieve accurate discrimination, we integrate Common Spatial Patterns (CSP) for spatial feature extraction along with Linear Discriminant Analysis (LDA) for classification. EEG recordings were obtained from 59 scalp electrodes positioned to capture neural activity in motor-related cortical regions. The signals were preprocessed using an 8–15 Hz band-pass filter to suppress noise and emphasize the sensorimotor rhythms relevant to movement imagination. Experimental evaluations on MI data demonstrate that the proposed CSP–LDA pipeline effectively distinguishes between different hand-imagery classes, suggesting its potential use in developing reliable and user-adaptive Brain–Computer Interface (BCI) systems for neurorehabilitation and assistive technology applications.

Index Terms—Electroencephalography, Motor Imagery, Brain–Computer Interface, Common Spatial Patterns, Linear Discriminant Analysis, Neurorehabilitation.

I. INTRODUCTION

The quest for seamless contact between humans and intelligent systems has expedited research in Brain–Computer Interfaces (BCIs) [1], which provide communication via cerebral impulses without requiring muscle exertion. BCIs convert brain activity into control commands capable of operating external devices [2], presenting revolutionary opportunities for individuals with significant motor impairments and for sophisticated human–machine interaction frameworks [3].

Motor Imagery (MI) [4], the imitation of movement in the mind without actual execution, has demonstrated significant potential among many BCI paradigms due to its capacity to elicit repeatable EEG signals from motor and premotor cortices. One can figure out what the user wants to do by looking at these brain oscillations, notably in the mu (8–13 Hz) as well as beta (13–30 Hz) frequency bands. Nevertheless, a significant problem in MI-based BCIs [5] is the precise differentiation of overlapping neural signatures, especially between left- and right-hand images, which exhibit strikingly analogous cortical activations [6].

Conventional BCI methodologies [7]–[9] frequently depend on constrained electrode arrangements or individualized calibration, potentially hindering generality and usefulness. To

address these constraints, this study employs a denser EEG montage and a feature-extraction methodology grounded in Common Spatial Patterns (CSP), integrated with Linear Discriminant Analysis (LDA) for resilient classification [9], [10]. The goal of this hybrid method is to make classes more distinct and make calculations easier, which will help create a better and more flexible BCI system. The study gives a thorough description of how to get data, clean it up, and use algorithms, with a focus on how possible it is to use CSP–LDA models for real-time MI identification and neuroprosthetic control [11], [12].

II. LITERATURE REVIEW

The development of Brain–Computer Interfaces (BCIs) [13], [14] has opened exciting possibilities for a more intuitive and direct form of human–computer interaction. By leveraging electroencephalography (EEG) to capture brain activity patterns, BCIs can potentially revolutionize the way individuals with motor impairments interact with the world. However, current BCI systems [15] face challenges in accurately classifying different motor imagery (MI) tasks, particularly for closely related movements like those involving the hands and feet. This necessitates further exploration of improved EEG data collection methods and the application of machine learning algorithms to enhance the classification accuracy of BCI systems. Wolpaw et al. [16] discuss Brain–Computer Interfaces (BCIs) as communication and control tools for individuals with paralysis. BCIs translate brain activity into commands, offering potential for those with ALS or brainstem stroke to operate software, control devices, or even participate in rehabilitation. The study explores various brain activity measurement methods like EEG, MEG, and fMRI. However, Wolpaw et al. highlight limitations of MEG, fMRI, and PET for everyday use due to cost, complexity, and real-time constraints. EEG is presented as a favorable option for BCIs due to its portability, affordability, and ability to provide real-time data.

Bajaj et al. [17] examined the use of EEG signals to categorize motor imagery (MI) tasks for brain–computer interfaces (BCIs) [18], [19]. Due to their non-invasive nature, EEG is

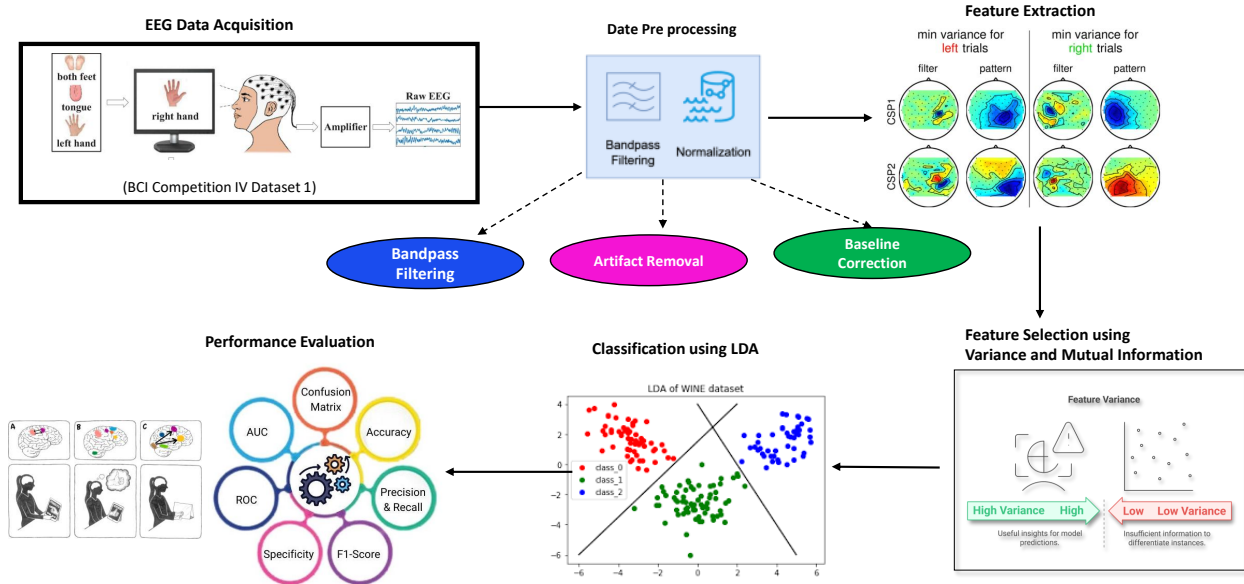


Fig. 1: Overview of the work

a preferred method for capturing brain activity related to imagined movements. The authors propose a method using RDWT (Rational Dilation Wavelet Transform) to decompose EEG signals into sub-bands, capturing oscillatory brain activity [20], [21]. Feature extraction is then performed to obtain time-domain features from these sub-bands. They evaluate these features with various machine learning algorithms, finding Quadratic Support Vector Machine (Q-SVM) [22], [23] to yield the best classification performance. This paper by Blankertz et al. [24] explores the Berlin Brain-Computer Interface (BBCI), a system designed for rapid and efficient communication between the brain and external devices. Unlike traditional BCI approaches that need extensive training, the BBCI uses machine learning to determine user intent from complex brain activity features. This greatly shortens the training time. A study showed that subjects could achieve control after just 20 minutes of calibration. The BBCI concentrates on users' motor skills and takes advantage of the brain activity linked to imagined movements. This method, along with the adaptability of machine learning algorithms, lets the system respond to individual user differences. The study found successful BCI control in most subjects, with information transfer rates over 35 bits per minute for some participants. Blankertz et al. [25] suggest that the BBCI's strength comes from its ability to utilize complex, relevant features and adjust to differences between users. This research leads to quicker and more user-friendly BCI technology.

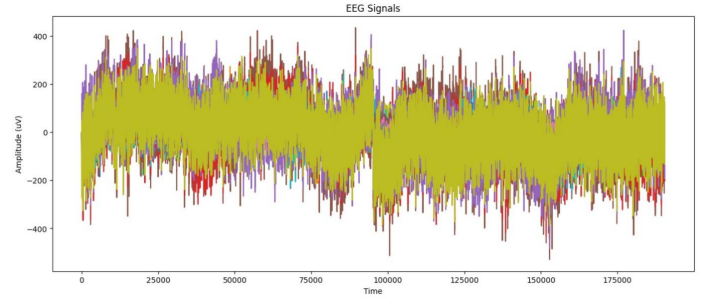
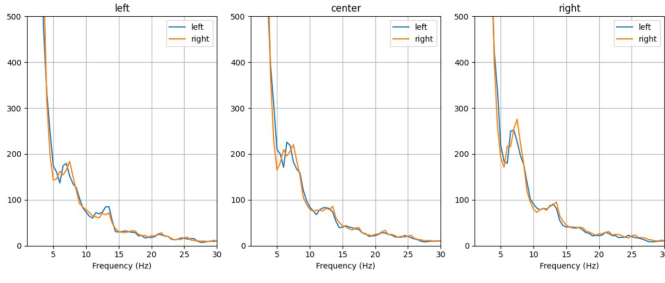


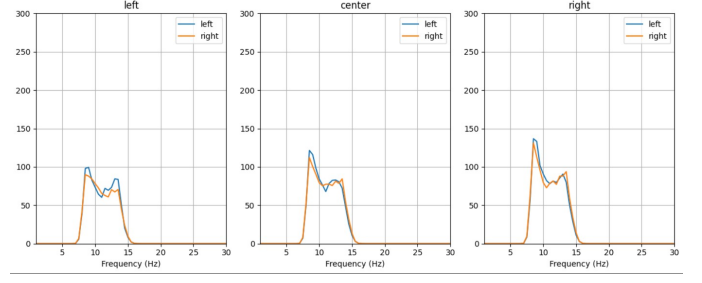
Fig. 2: Raw EEG Signals before denoising

III. PROBLEM STATEMENT

Millions of individuals worldwide are struggling with motor impairments caused by conditions like sudden stroke, spinal cord injury, or Amyotrophic Lateral Sclerosis (ALS). These disease significantly restrict their ability to interact with the environment, hindering independence and overall well-being. State of art assistive technologies, while valuable, may have drawbacks in terms of functionality, ease of use, or invasiveness. This research project seeks to address this challenge by exploring the potential of Brain-Computer Interfaces (BCIs) for a more intuitive and direct form of human computer interaction. Our specific focus lies in developing a system capable of classifying motor imagery (MI) tasks associated with motor imagery of right and left-hand movements. The primary problem lies in accurately recognising the subtle variations in

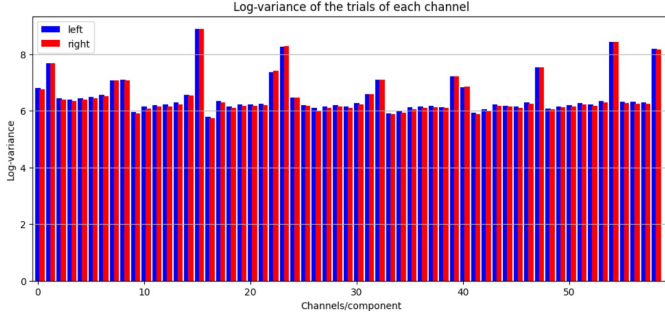


(a) Before filtering the signal

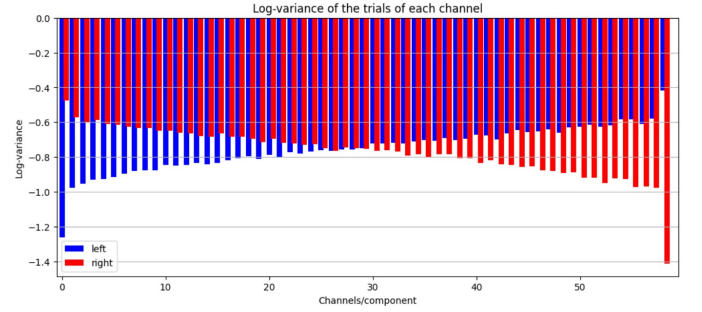


(b) After Applying BandpassFilter

Fig. 3: Power spectral density (PSD) of EEG signals for channels C3, Cz, and C4, calculated for trials



(a) Before CSP



(b) After CSP

Fig. 4: Log-Variance of the trials of each channels

EEG signals generated during the imagination of right and left-hand movements. These closely related motor representations often exhibit overlapping brain activity patterns, posing a significant challenge for existing BCI systems. Misclassifications can lead to frustration and hinder the user experience. This project aims to overcome this limitation by investigating the effectiveness of Machine Learning (ML) algorithms [26] in analyzing EEG data collected during real and imagined hand movements. By employing advanced feature extraction and classification techniques, we aim to develop a BCI system that can accurately decode user intent based on their motor imagery tasks.

A. Methodology

To make a strong BCI system for classifying hand movement images, we need to plan and carry out the work very carefully. Here are the main steps that need to be taken:

1) *Data Acquisition: Data Description:* We have tested our mechanism with publicly available EEG datasets like BCI Competition IV Dataset 1.

2) *Data Preprocessing and Feature Engineering:* The raw EEG data has undergone preprocessing steps to remove noise and artifacts. This may include filtering (band-pass filtering between 0.05-200 Hz to isolate sensorimotor rhythms), baseline correction, and artifact removal techniques like Independent Component Analysis (ICA) to address eye blinks or muscle activity.

3) *Feature Extraction::* Techniques like Common Spatial Patterns (CSP) [27] have been employed to extract the most

discriminative features from the preprocessed EEG data. CSP identifies spatial filters that maximize the variance of EEG signals associated with specific MI tasks (left vs. right hand) while minimizing noise and signals from other tasks. The Common Spatial Pattern (CSP) algorithm seeks spatial filters that increase variation in EEG signals for one class while minimizing it in another.

Let $X_i \in \mathbb{R}^{C \times T}$ denote the EEG data matrix for class i , where C is the number of channels and T is the number of time samples. The normalized covariance matrix is computed as:

$$C_i = \frac{X_i X_i^T}{\text{trace}(X_i X_i^T)} \quad (1)$$

The composite covariance matrix is given by:

$$C = C_1 + C_2 \quad (2)$$

CSP seeks a projection matrix W that diagonalizes both C_1 and C_2 simultaneously:

$$W^T C_1 W = \Lambda_1, \quad W^T C_2 W = \Lambda_2 \quad (3)$$

where Λ_1 and Λ_2 are diagonal matrices containing the eigenvalues.

The spatially filtered signals are obtained as:

$$Z = W^T X \quad (4)$$

Finally, the log-variance feature for each channel is com-

Algorithm 1 Common Spatial Pattern (CSP) Feature Extraction

Require: EEG signals X_L, X_R for left and right hand classes

Ensure: Projection matrix W and CSP feature vector f
Compute normalized covariance matrices:

$$C_L = \frac{X_L X_L^T}{\text{trace}(X_L X_L^T)}, \quad C_R = \frac{X_R X_R^T}{\text{trace}(X_R X_R^T)}$$

Compute composite covariance matrix:

$$C = C_L + C_R$$

Solve the generalized eigenvalue problem:

$$C_L W = \lambda C W$$

Select eigenvectors corresponding to largest and smallest eigenvalues.

Project EEG signals: $Z = W^T X$

Compute log-variance features:

$$f_j = \log \left(\frac{\text{var}(Z_j)}{\sum_k \text{var}(Z_k)} \right)$$

return f as CSP feature vector. =0

puted as:

$$f_j = \log \left(\frac{\text{var}(Z_j)}{\sum_k \text{var}(Z_k)} \right) \quad (5)$$

These log-variance features are used as input to the classifier.

2. Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) looks for a linear combination of features that best distinguishes two or more classes. Let μ_1 and μ_2 denote the mean vectors of two classes, and S_1, S_2 their within-class scatter matrices. The within-class scatter matrix S_W and between-class scatter matrix S_B are given by:

$$S_W = S_1 + S_2, \quad S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \quad (6)$$

The optimal projection vector w maximizes the Fisher criterion:

$$J(w) = \frac{w^T S_B w}{w^T S_W w} \quad (7)$$

The solution for w is obtained as:

$$w = S_W^{-1}(\mu_1 - \mu_2) \quad (8)$$

A new sample x is classified using the discriminant function:

$$y = w^T x + b \quad (9)$$

where b is the bias term derived from training data.

4) *Feature Selection*:: From the features extracted by CSP, a subset of the most informative features relevant for classification has been selected using techniques like feature variance or mutual information [28]. This helps to reduce computational complexity and potentially improve classification accuracy [29].

Algorithm 2 Linear Discriminant Analysis (LDA) Classification

Require: Feature vectors f_i and corresponding labels y_i

Ensure: Projection vector w and predicted class $\hat{y}$ Compute class means:

$$\mu_1 = \text{mean}(f_i | y_i = 1), \quad \mu_2 = \text{mean}(f_i | y_i = 2)$$

Compute within-class scatter:

$$S_W = S_1 + S_2$$

Compute between-class scatter:

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

Derive optimal projection:

$$w = S_W^{-1}(\mu_1 - \mu_2)$$

For each test vector x , compute discriminant score:

$$y = w^T x + b$$

Classify sample:

$$\hat{y} = \begin{cases} 1, & \text{if } y > 0 \\ 2, & \text{otherwise} \end{cases}$$

return $\hat{y}$ as predicted class. =0

5) *Machine Learning and Classification*:: Machine learning algorithms like Linear Discriminant Analysis (LDA) or Support Vector Machines (SVM) [30] have been implemented for classification. These algorithms learn to distinguish between the EEG patterns associated with left and right-hand motor imagery tasks based on the extracted features. The BCI system [31] has been trained using a portion of the preprocessed data. The trained model has then been evaluated on a separate unseen portion of the data to assess its generalization ability and classification accuracy. The performance of the chosen machine learning algorithms [32] has been further optimized by tuning their hyperparameters (e.g., regularization parameters in an SVM). Techniques like grid search or randomized search have been used to identify the optimal hyperparameter configuration [33].

B. Experiment and Result

The overall performance of the BCI system has been evaluated through user studies, measuring factors like classification accuracy, information transfer rate (ITR), and user satisfaction. Based on the evaluation results, the BCI system has been iteratively refined. This may involve adjusting data acquisition protocols, feature extraction techniques, or machine learning algorithms to enhance the system's performance and usability.

Having extracted informative features through techniques like bandpass filtering [34], we now delve into the crucial stage of model preparation for our BCI system. This section explores various methods to equip the model for effectively classifying left and right hand movement imagery tasks based

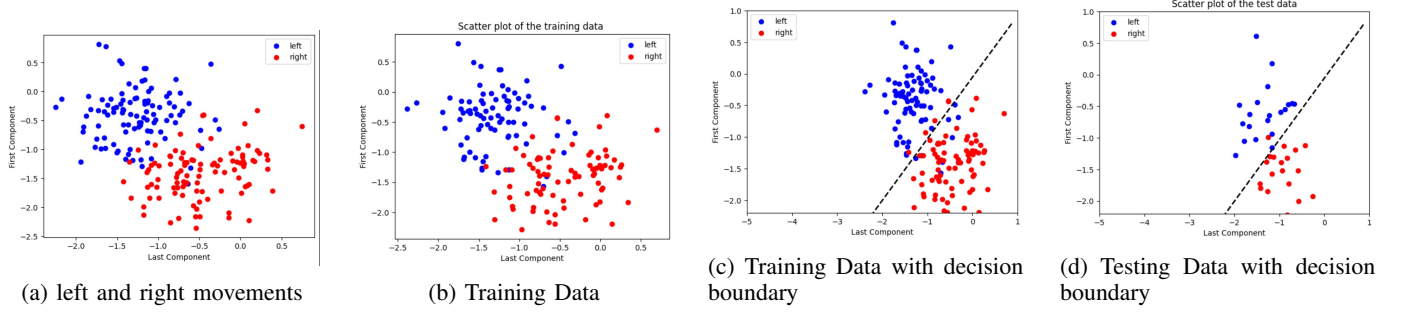


Fig. 5: Comparison between the scatterplot

on the extracted features. We will explore techniques like log-variance calculation, Common Spatial Patterns (CSP), and Linear Discriminant Analysis (LDA) to enhance feature discrimination and ultimately achieve robust classification performance.

1) *Log Variance Calculation*: One approach to feature extraction involves calculating the log-variance of the EEG signals. The provided code implements a function named `logvar` that computes the log-variance across time samples for each channel within a trial. This essentially reflects the variability or spread of the signal's amplitude within a specific channel over time.

a) *Plotting the PSD after applying CSP*: Figure 4 shows the Power Spectral Density graph plotted after applying filters and CSP:

b) *Scatterplot of the trials*: Following the application of CSP filters, this section explores the resulting features using a scatter plot for visualization (Figure 5).

C. Testing and Result Analysis

This section focuses on evaluating the performance of the trained LDA model [35] on unseen data. The provided confusion matrix shows the classification results for the test data: 20, 1 : 20 trials from the left-hand class were correctly classified as left hand, and 1 trial from the right-hand class was mistakenly classified as left hand (false positive). 0, 19 : 0 trials from the left-hand class were mistakenly classified as right hand (false negative), and 19 trials from the right-hand class were correctly classified as right hand. The accuracy is calculated as 0.975, indicating that the model correctly classified 97.5% of the trials in the testing data set. This suggests that the model has learned effective features and decision boundaries that generalize well to unseen data. Figure shows the ROC curve to visualize the performance of a binary classifier at different classification thresholds.

IV. CONCLUSION

This study presented an adaptive EEG-based motor imagery classification framework integrating Common Spatial Pattern (CSP) feature extraction with Linear Discriminant Analysis (LDA) for robust Brain-Computer Interface (BCI) applications. By employing efficient spatial filtering and discriminative linear modeling, the proposed CSP-LDA pipeline

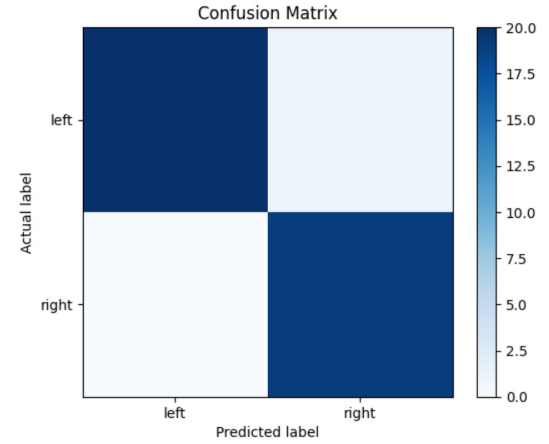


Fig. 6: Confusion Matrix of the Trials

achieved reliable separation between left- and right-hand motor imagery tasks, attaining an accuracy of approximately 97.5%. The results demonstrate the capability of combining spatial filtering and statistical learning to enhance EEG signal interpretability and classification performance. This approach contributes to the development of user-adaptive, non-invasive BCI systems for neurorehabilitation and assistive control. Future research will focus on extending this work to multi-class motor imagery, incorporating deep learning-based spatial filters, and validating the framework on larger and more heterogeneous datasets to further improve real-time adaptability and generalization.

DECLARATION ON GENERATIVE AI

During the preparation of this article, the author(s) used Grammarly in order to check grammar and spelling. The author(s) take(s) full responsibility for the article's content.

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